

A Transformer-Based Framework for Automated Product Rating Generation and Inconsistency Detection in E-Commerce Reviews

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Abstract

Customers of e-commerce platforms heavily rely on user-generated product reviews for decision making. However, these reviews often suffer from rating inconsistencies as well as biased evaluations, which can mislead customers, reduce trust in review systems and undermine trustworthy decision-making. Consequently, our study found significant discrepancies between the displayed product ratings and the sentiments expressed in customer reviews. This study proposes a novel approach for generating product ratings on e-commerce platforms by automatically deriving ratings from sentiment-driven customer reviews rather than relying on manual rating. Using mixed-language reviews from a mid-sized dataset, the framework analyzes review sentiments, predicts product ratings and identifies inconsistencies between sentiment driven rating and platform given manual rating. Our proposed framework leverages pre-trained transformer model BART (used in a zero-shot setting for sentiment detection), and adopting a five-class sentiment scheme (very positive, positive, neutral, negative, very negative) and transformer base models on the dataset, this study captures nuanced sentiments more effectively than conventional three-class approaches, achieving strong and reliable rating. Specifically, DeBERTa, RoBERTa are highly optimized transformer architecture on our custom dataset to enhance performance and adapt to the task of sentiment-driven rating prediction. The sentiment classification model achieved 89% accuracy, while the proposed rating prediction framework obtained an MAE of 0.15 when comparing sentiment-derived ratings with actual user ratings. Comparative analysis demonstrates that transformer-based models significantly outperform traditional baseline methods. Overall, the findings indicate that the proposed framework enhances review reliability, supports platform moderation and improves user trust and decision-making on e-commerce platforms.

Keyword: E-commerce, Inconsistencies, Transformer Models, Sentiment-driven Rating, Decision-Making, Reliability

1. Introduction

E-commerce growth has reshaped how consumers evaluate products online, with digital platforms now mediating purchase decisions previously made through physical inspection. User-generated product reviews have emerged as the most influential factor shaping purchase decisions, with over 90% of online shoppers consulting reviews before buying. These reviews serve as virtual word-of-mouth, bridging the information gap where physical product examination is impossible [1].

However, rating inconsistencies and biased evaluations have become pervasive, where numerical star ratings contradict textual sentiment or fail to reflect true satisfaction, ultimately misleading customers. For instance, reviewers may assign 5 stars while describing defects, or 1 star despite expressing satisfaction. These discrepancies arise from cultural differences in rating interpretation, linguistic variation, deliberate manipulation, cognitive biases, or interface misunderstanding [2-3].

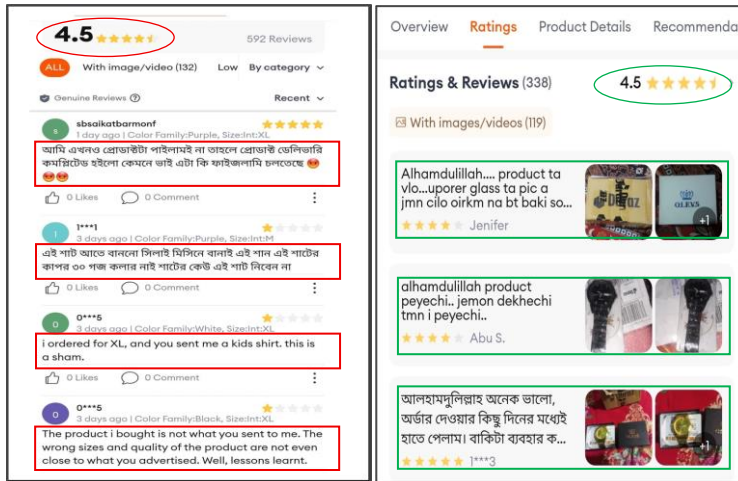


Fig. 1: Examples of rating-review inconsistency in two e-commerce products

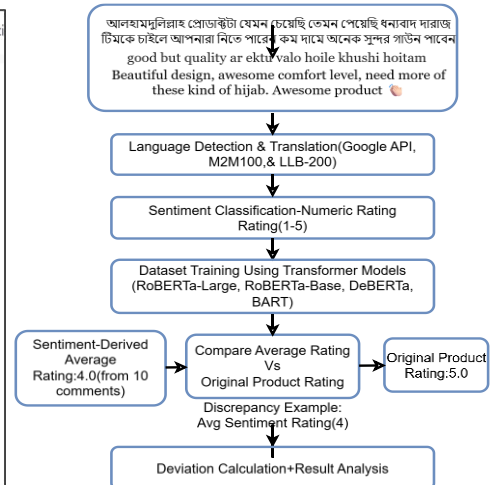


Fig.2: Proposed process workflow

Fig. 1 shows two different product overviews where both products have the same high ratings, but their reviews say that there are different customer experiences. In the first case, with the high rating, the reviews contain significantly negative feedback and dissatisfaction with the product. On the other hand, the second product has a high rating that is supported by positive customer reviews. This scenario shows a rating and review inconsistency, where the displayed rating does not accurately align with the sentiment expressed in the reviews. These types of inconsistencies occur because of external influences, including incentivized ratings, manipulated reviews, or third parties connected to attempts at manually increasing product ratings. When customers rely on ratings as a quick indicator of product quality, many may purchase a product without carefully checking the reviews. Therefore, they may receive products that do not meet expectations, leading to dissatisfaction when the product's showing a high rating.

Such inconsistencies systematically erode consumer confidence, lead to suboptimal purchase decisions, and degrade recommendation algorithm performance. When aggregated ratings fail to reflect true sentiment, both consumers and sellers suffer from misaligned expectations. Maintaining review quality is therefore critical for platform competitiveness and user retention.[5][6] Traditional sentiment analysis methods including rule-based approaches, lexicon scoring, Support Vector Machines, and Naive Bayes classifiers have shown limited success with real-world review complexity. They struggle with sarcasm, domain-specific language, multilingual text, and nuanced sentiment gradations beyond simple positive-negative classifications. Simplified three-class schemes compress human emotional expression, resulting in information loss. In contrast, a five-class scheme such as very negative, negative, neutral, positive, very positive for better captures emotional nuance [6][7]. Transformer-based models have addressed many of these limitations. BERT enabled bidirectional contextual understanding, while RoBERTa and DeBERTa-V3 (micro) refined performance through improved training strategies. BART-BASE, by contrast, demonstrates strong zero-shot capabilities pre-training applicable to this domain. Specifically, Abumalloh et al. demonstrated that transfer learning substantially improves classification performance in low-resource multilingual settings, including code-mixed text.[8-12] Despite these advances, critical gaps remain: coarse sentiment classification masks emotional nuances, systematic inconsistency detection frameworks are underdeveloped, and practical real-time deployment challenges in code-mixed environments are largely unaddressed [4][3].

For the making bridge between review rating, fig. 2 show this study proposes a transformer-based framework for sentiment-driven rating estimation and inconsistency detection. The key novelty lies in decoupling star ratings from text sentiment through logical, context-aware mechanisms validated on code-mixed Bangla-English reviews, addressing quality control challenges in emerging e-commerce platforms. Evaluated on a mixed-language dataset of Bangla, English, and code-mixed e-commerce reviews, the framework employs BART-BASE in zero-shot mode and fine-tuned DeBERTa-V3 (micro), RoBERTa-base, and RoBERTa-large. It introduces fine-grained five-class classification, multi-model comparative analysis across four transformer architectures, and automated quality control mechanisms, achieving 89% accuracy in rating prediction and 0.15 mean absolute error in inconsistency detection. The framework operates through three core phases, as illustrated in Figure 2, each addressing a distinct stage of the review analysis pipeline.

The main contributions are: (1) A localized framework for processing code-mixed e-commerce reviews that detects mismatches between review text and star ratings; (2) A comparative evaluation of four transformer models for five-class sentiment analysis in multilingual settings; (3) Groundwork for logical rating systems that operate independently

of text sentiment to restore customer trust. The rest of the paper is organized as follows: Section 2 reviews related work, Section 3 details the methodology, Section 4 presents experimental results, and Section 5 concludes the paper with future directions.

2. Literature Review

One of the most popular current research directions in the intersection of natural language processing (NLP), machine learning, and e-commerce analytics is sentiment analysis of product reviews. The dynamic development of online shopping services has resulted in massive quantities of customer feedback that offers useful information on customer perceptions and buying patterns [1], [5]. These reviews are greatly related to product satisfaction and purchases and, thus, automated sentiment analysis is critical to comprehend customer reviews and aid decision-making in business [4], [6], [7]. Sentiment analysis Methodologically, sentiment analysis has developed out of the more traditional machine learning methods to deep learning models and more recently large language models (LLMs) to do so in a more contextual and accurate manner [20], [30]-[32]. The emerging literature will show that sentiment analysis is not only useful in predicting product rating but also fine-grained opinion mining, trend analysis, and recommender systems that offer personalized recommendations [19], [21], [24], [29], [30], [42]. This review provides an overview of the latest developments in sentiment analysis of reviews of products, including methodological aspects, most frequently utilized datasets, evaluation protocols, and shortcomings of the current research.

2.1 Classical Machine Learning and Ensemble-Based Methods.

Sentiment analysis of product reviews has been extensively done using traditional machine learning methods. Yadav and Peddinti [25] compared SVM, Naive Bayes and Random Forest based classification with Amazon reviews and had accuracy ranging between 82 percent and 87 %, where SVM was the best. Ullah et al. [30] obtained an accuracy of 84.3% on electronics reviews with Random Forests and Sethi and Gupta [32] focused on the role of preprocessing and feature engineering, observing a weakness in semantic understanding. Ensemble and optimization-based methods have been proposed to improve performance. The team by Mudarakola et al. [27] got an accuracy of 88.2% with an optimized ensemble framework, whereas Godia and Tiwari [16] had an accuracy of 89.5 % with ensemble models on Amazon reviews, but issues of sarcasm, domain specificity, and multilingual data still remained. With such developments, the traditional and ensemble-based approaches have the disadvantage of having poor contextual modeling, manual feature engineering, and sensitivity to class imbalance, which has prompted the implementation of deep learning methods [31].

2.2 Deep Learning Sentiment Analysis Methods.

Deep learning models have significantly enhanced automated feature learning and contextual modelling of sentiment analysis of product reviews. Recurrent models like LSTM and BiGRU have demonstrated good performance with Atlas et al. [22] achieving a 91.7 % accuracy with BiGRU on Amazon reviews and Sakib et al. [26] with LSTM-based models. More accurate results of 92.4% are further obtained using hybrid CNNLSTM frameworks that have integrated local and sequential feature extraction but with more complexity [26], [29]. Transformer-based models, especially BERT and its variations, have recorded state-of-the-art performance. Maximum accuracies of domain-adapted and attention-enhanced BERT models were 94.6% indicating the efficacy of transfer learning and self-attention systems [8-10], [36]. Transformer-recurrent architectures with hybrid transformers performed the best with Bellar et al. [29] recording a 95.1% accuracy. These models are demanding in terms of large datasets and high computing capabilities even though they are better performing [43].

2.3 Sentiment Analysis using a Large Language Model.

Large language models (LLMs) are the most recent model in product review sentiment analysis, which has better contextual and semantic comprehension. Compared to traditional and deep learning algorithms, GPT and LLaMA models are found to provide better results in comparative studies, and GPT-4 report higher than 96 percent accuracies in multi-platform product review datasets [17]. Ghatora et al. [15] also showed that GPT-3.5 has good performance in zero-shot and few-shot tasks and underline the flexibility of LLMs without fine-tuning on specific tasks, but the cost of inference and scalability are still issue. LLM adoptions specific to domains and language have demonstrated specific success in low-resource environments. Multilingual e-commerce reviews of the Bengali and Bangali language with fine-tuned LLMs produced 91.8 to 93.6 accuracy, which validates that domain-sensitive and language-specific training can enhance its performance despite the limited amount of data and challenges of computation [12-14], [28], [43]. In addition to document-level classification, LLMs have also been used to implement more advanced applications, including the ability to extract sentiment at the sub-feature level and a personal recommendation system. GPT-4 based models were found to be considerably more accurate in fine-grained opinion mining and yield better results in recommendation through an increase of more than 18 percent but at the cost of prompt engineering complexity, real-time processing and resource consumption is a major drawback [17], [19], [29], [30].

3. Methodology

The proposed system architecture follows a four-stage pipeline for sentiment-driven product rating generation and inconsistency detection in fig.3. First, product review data are collected from both manually gathered and publicly available sources, then cleaned through preprocessing steps such as noise removal, language detection, translation, and

normalization. Second, the cleaned reviews are annotated using a hybrid labeling strategy that combines manual expert labeling with BART-based zero-shot auto-labeling to assign five sentiment classes from very negative to very positive.

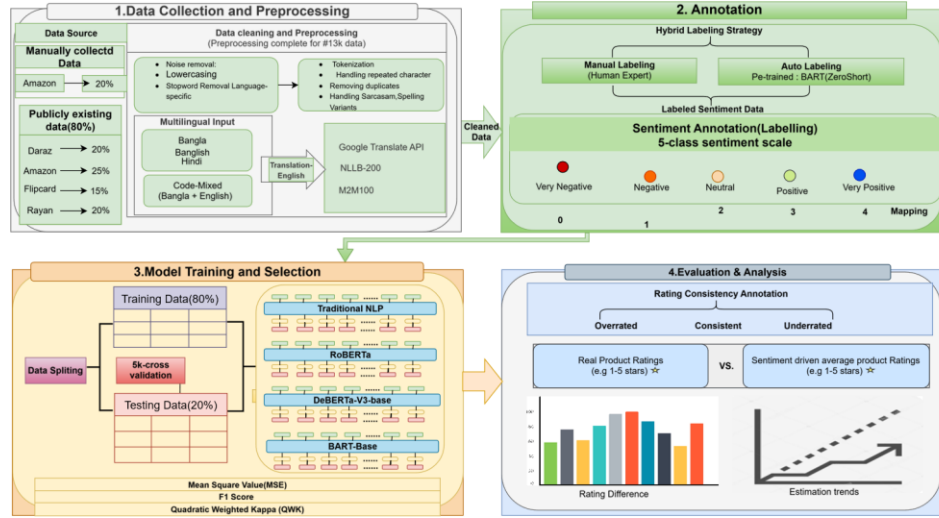


Fig. 3 System architecture

After annotation, the dataset is used to train and evaluate multiple models, including traditional machine learning classifiers and transformer-based models such as BART, RoBERTa, and DeBERTa. Finally, the best-performing model generates sentiment-derived ratings, which are compared with platform-provided ratings. This comparison enables the system to classify products as overrated, consistent, or underrated, supporting review reliability analysis and helping detect rating-review inconsistencies in e-commerce platforms.

3.1 Dataset Description

Data for this research was collected using a mixed-source approach, with 20% from manual methods and 80% from publicly available datasets to some of feature trends collected. The final dataset contains 13,620 e-commerce product reviews. Of these, 10,896 reviews were collected from publicly available sources, while 2,724 reviews were manually collected from Bangladeshi e-commerce platforms. Each record includes SKU, product category, review text, user-assigned review rating, platform product rating, language type, and sentiment label. illustrated in Table 1.

3.2 Data Annotation

A hybrid annotation strategy was used. First, 30% of the dataset was manually annotated by domain-aware human annotators. The remaining 70% was labeled using a BART-based zero-shot classifier. To reduce labeling noise, BART predictions with confidence below $\tau = 0.60$ were manually reviewed. A subset of 500 samples was independently labeled by two annotators to measure inter-annotator agreement using Cohen's kappa. Inter-annotator agreement will be included in future dataset validation.

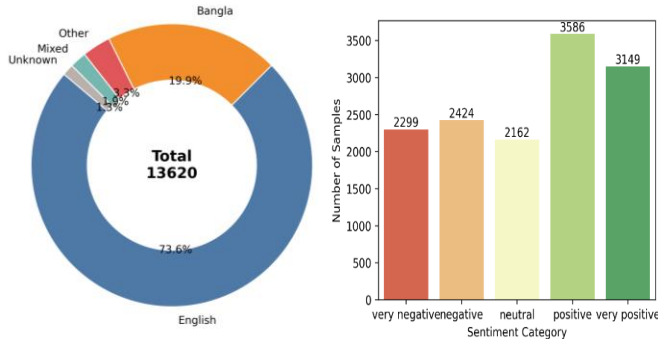


Fig. 4 Language variation and per classes review

In fig. 4 illustrate that dataset includes 2,299 very negative, 2,424 negative, 2,162 neutral, 3,586 positive, and 3,149 very positive comments. The collection of data carefully handle the balance with each sentiment category that effect our model accuracy. In this research, multiple classifiers use for classifier an English, Bangla, Banglish classifier were used to categorize languages.

3.3 Language Detection & Translation

The findings, shown in Fig. 3, indicate that English is the predominant language for online product reviews by Bangladeshis, comprising 73.6% of all reviews. Other significant languages include Bangla at 19.9%, mixed at 1.3%, and others or mixed at 3.3%, with 1.9% classified as unknown (Banglish). All non-English and code-mixed reviews were translated into English before model training. Three translation engines were compared: Google Translate API, NLLB-200, and M2M100. Each translated version was passed through the same RoBERTa-Large classifier on the held-out set. Since Google Translate achieved the highest downstream sentiment performance, it was selected as the final translation pipeline for the main experiments[11-13].

3.4 Data Preprocessing

Each record undergoes two preprocessing phases: Phase 1 (noise reduction) lowercasing, stopword removal, and removal of special characters/HTML artefacts; Phase 2 (normalization) tokenization, repeated-character collapsing (e.g., 'greaaaat' → 'great'), duplicate removal, and standardization of spelling variants and sarcasm markers. Preprocessing was applied uniformly across the full corpus. [9-10]

3.5 Model Selection and Configurations

Two experimental approaches were used: (1) zero-shot sentiment detection using BART-based zero-shot classifier, and (2) fine-tuning of BART-Base, DeBERTa-V3-Base, RoBERTa-Base, and RoBERTa-Large on the ~13k dataset. The fine-tuning stack comprised Transformers 4.38.2, Accelerate 0.28.0, and PEFT 0.8.2. The dataset was divided using stratified sampling into 80% training data and 20% held-out test data. The training portion was further validated using five-fold cross-validation during model selection. Final performance was reported only on the independent 20% test set. The

Table 1. Dataset Size and Distribution by Source

Source	Manual/ Exiting Dataset Reviews	Language
Daraz	1000/5521	Bangla/mix
Amazon	520/3005	English
Flipkart	400/1340	English/Mix
Ryans	804/1030	English
Total: 13,620 Reviews		

unified training configuration applied across all models was: maximum token length 128, training epochs 5, batch size 8 per device, learning rate 2e-5, weight decay 0.01 (L2 regularisation), and warmup ratio 0.1.

3.6 Sentiment Classification

In table 2 shows all records are annotated using a five-class ordinal sentiment taxonomy, essential for mapping sentiment predictions onto a 1–5 star scale from Eq.(1):

Table 2. Five-Class Ordinal Sentiment Taxonomy

Class	Integer	Defining Characteristics
Very Negative	1	Severe dissatisfaction; product failure; explicit anger; return demands
Negative	2	General disappointment; unmet expectations; moderate criticism
Neutral	3	Factual or balanced; mixed opinions; no dominant polarity
Positive	4	Satisfaction; moderate enthusiasm; implied recommendation
Very Positive	5	Strong enthusiasm; superlatives; explicit repeat-purchase intent

The final sentiment label is assigned by selecting the class with the highest predicted probability: $s_i = \text{argmax}_{c \in C} P(c | x_i)$, where x_i is the i -th review, C is the set of sentiment classes, and s_i is the predicted label.

3.7 Inconsistency Detection Mechanism

This mechanism identifies cases where aggregate review sentiment diverges significantly from official product ratings potentially indicating rating manipulation, cognitive bias, or cultural misinterpretation of the rating scale. For each product p comprising n reviews, the sentiment-driven predicted star rating $\hat{R}_p$ is computed as a probability-weighted average[16][17]:

$$\hat{R}_p = \frac{1}{n} \sum_{i=1}^n \sum_c P(c | r_i) s_c \quad (1)$$

where $P(c | r_i)$ is the model-predicted probability for class c given review r_i , and where $s_c \in \{1,2,3,4,5\}$ is the star score assigned to sentiment class c .

Products with fewer than 20 reviews are excluded from inconsistency analysis (Lines 3–5) to reduce the impact of small-sample bias. The sentiment-driven average rating is computed and compared with the platform rating (Lines 8–10). Based on the rating difference, products are categorized as Consistent, Overrated, or Underrated (Lines 12–17). In Eq. (2), N_p denotes the number of reviews for product p , and r_i denotes the sentiment-derived rating of the i -th review. The difference between the platform rating and the sentiment-driven rating is then calculated as:

Algorithm 1: Product-Level Rating Inconsistency Detection

Input:	7: $SR_p \leftarrow \text{Mean}(\text{Sentiment Ratings})$
P : Product set	8: $PR_p \leftarrow \text{Platform Rating}$
T : Minimum review threshold (=20)	9: $D_p \leftarrow PR_p - SR_p$
θ : Consistency threshold (=0.3)	10: if $ D_p < \theta$ then
Output:	11: $C_p \leftarrow \text{Consistent}$
Output: C_p : Consistency category for	12: else if $D_p > \theta$ then
each product p	13: $C_p \leftarrow \text{Overrated}$
1: for each product $p \in P$ do	14: else
2: $N_p \leftarrow \text{Number of reviews}$	15: $C_p \leftarrow \text{Underrated}$
3: if $N_p < T$ then	16: end if
4: $C \leftarrow \text{Insufficient Reviews}$	17: end for
5: continue	18: return C_p
6: end if	

$$\text{Sentiment-Driven Average Rating: } SR_p = \frac{1}{N_p} \sum_{i=1}^{N_p} r_i \quad (2)$$

where PR_p denotes the platform-provided product rating. Finally, the product consistency category C_p is assigned according to the following rule:

$$\text{Rating Difference: } D_p = PR_p - SR_p \quad (3)$$

This categorization is applied only when the product satisfies the eligibility constraint:

Consistency Categorization: $\theta = 0.3$

$$C_p = \{\text{Consistent}, |D_p| < \theta \text{ Overrated}, D_p > \theta \text{ Underrated}, D_p < -\theta\} \quad (4)$$

$$\text{Eligibility Constraint: } N_p \geq 20 \quad (5)$$

Algorithm 1 is supported by four mathematical definitions. First, for each eligible product p , the sentiment-driven average rating SR_p is calculated by averaging the sentiment-derived ratings r_i across all N_p reviews. Next, Eq.(3) the rating difference D_p is obtained by subtracting SR_p from the platform-provided rating PR_p . This difference is then used to assign the product into one of three consistency categories Eq.(4): Consistent, Overrated, or Underrated. To reduce small-sample bias, the categorization is performed only for products with at least 20 reviews in Eq. (5).

3.8 Evaluation Metrics

Mean Absolute Error (MAE) is used to measure the average absolute difference between the predicted rating and the actual rating. n is total number of samples $\hat{y}_i$ is predicted rating/sentiment score for sample y_i is actual rating/sentiment score for sample i . Since the task uses ordinal rating values, MAE is suitable because it directly reflects how far the predicted score deviates from the true score on average [18].

$$MAE = \frac{1}{n} \times \sum_{i=1}^n |\hat{y}_i - y_i| \quad (6)$$

Macro-averaged F1 assesses per-class precision-recall balance, treating each class equally regardless of support size. For class c : [19]

$$Prec^c = TP^c \div (TP^c + FP^c) \quad (7)$$

$$Rec^c = TP^c \div (TP^c + FN^c) \quad (8)$$

$$F1^c = 2 \times Prec^c \times Rec^c \div (Prec^c + Rec^c) \quad (9)$$

$$F1_{macro} = \left(\frac{1}{c}\right) \times \sum^c F1^c \quad (10)$$

Quadratic Weighted Kappa (QWK) is the primary model selection metric because it explicitly penalises larger ordinal label discrepancies. The weight for a disagreement between true class i and predicted class j is [15][20]:

$$\kappa = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}} \quad (11)$$

Where:

- O_{ij} = observed agreement matrix
- E_{ij} = expected agreement matrix
- w_{ij} = quadratic weight

$$\text{Quadratic weight: } w_{ij} = \frac{(i-j)^2}{(k-1)^2} \quad (12)$$

Where:

- $i, j \in \{1, 2, 3, 4, 5\}$
- $k = 5$ rating categories
- Ratings are ordinal
- It penalizes large prediction differences more heavily

Table 3 summarizes the key equations and evaluation metrics used to assess sentiment-driven rating prediction and rating inconsistency. It includes MAE, F1-score, Quadratic Weighted Kappa, and sentiment-derived product rating measures.

Table 3. Summary of Equations and Evaluation Metrics

Equation	Metric	Purpose
(1)–(5)	Inconsistency	Sentiment driven rating and platform rating
(6)	MAE	Ordinal-sensitive prediction error
(7)–(10)	F1 (macro)	Class-balanced classification quality
(11)–(12)	QWK	Primary ordinal-aware model selection metric
(1)	$\hat{R}_p$	Sentiment-driven product star rating
(3)	ΔR_p	Rating consistency classification basis

4. Results Analysis

This study was conducted through two separate experiments, one part is focusing on sentiment detection, on the other part finding the inconsistency of existing e-commerce system. The product “Rating” typically carry the quantity of user like a product after buying it and The “Review” field explains why they like or dislike it of detailed descriptions of these comments. Then collect large quantity of balance class dataset from Bangladesh e-commerce shops and some of existing dataset to specific feature. These features are chosen based on their significance and nature of the problem. These experiments were designed to train, test and evaluate the effectiveness of two distinct approaches pre-training and training LLMs. The objective was to Sentiment-driven rating comparison for find inconsistency between rating and review system in e-commerce site. Throughout, performance is quantified using Accuracy, Macro-F1, Weighted-F1, Mean Absolute Error (MAE), and Quadratic Weighted Kappa (QWK), with MAE capturing average rating deviation and QWK penalizing larger ordinal disagreements more heavily.

4.1 Comparative Analysis of Translation Models

Because the corpus contains reviews written in Bangla, Banglish (romanized Bengali), Hindi, and code-mixed (Bangla + English) variants, a reliable translation pipeline is a prerequisite for downstream sentiment classification. Three translation services were benchmarked: Google Translate API, NLLB-200, and M2M100. Table 4 provides representative qualitative outputs for each input variety, while Table 5 reports quantitative metrics derived from the downstream sentiment-classification task performed on each system's translated output.

Table 4. Qualitative translation outputs for Bangla, Banglish, and code-mixed review samples.

Original Text (Bangla)	“কাপড় একদম ভালো না । জামা ছিঁড়া ছিল। ছবির সাথে জামার কোনো মিল নেই”
Google API	“The clothes are not good at all. The clothes were torn. The clothes do not match the picture.”
NLB 200	“The clothes weren't very good. The clothes were torn.”
M2M 100	"Cloth is not good at all. Shirt was torn. There is no match of the shirt with the picture."
OriginalT ext (Banglish)	“Good product but price aktu beshi lagce amer but this is best,,hight a aktu boro hole vlo hoto”
Google API	“Good product but price is a bit high but this is the best,,hight a price boro hole vlo hoto”

NLB 200	"Good product but price aktu beshi lagce amer but this is best,,Hight an aktu boro hole vlo hoto"
M2M 100	"Good product but price aktu beshi lagce amer but this is best,,hight a aktu boro hole vlo hoto"
Original Text (Mixed)	"মোটা মটি ভালো prise hisebi tik ase chole kapor ta r akto valo hyle hyto"
Google API	"It's average. Considering the price, it's okay and passable. It would have been better if the fabric was a little bit higher quality."
NLB 200	"I'm not sure if I'm going to be able to do this."
M2M 100	"মোটা মটি ভালো prise hisebi tik ase chole kapor ta r akto valo hyle hyto"

4.2 Quantitative Impact of Translation Quality on Downstream Classification

To objectively measure translation effectiveness, each system's output was fed into the best-performing classifier and evaluated on the same held-out set. Table 5 summarizes the results. The Google Translate API achieved the best results with 89.02% accuracy, 0.902 micro-F1, 0.151 MAE, and 0.939 QWK. NLLB-200 obtained moderate performance, while M2M100 produced the lowest scores across all evaluation metrics. The result directly affects the quality of sentiment analysis performance. Where Google Translate identifies the accurately reviewed language, like Bangla, Banglish, and mixed, it then converts it into English, which is more effective. That's why it was chosen as the language conversion medium for further experiments.

Table 5 Sentiment classification performance as a function of the translation pipeline used.

Translation Model	Accuracy	Micro-F1	Weighted-F1	MAE	QWK
M2M100	0.6569	0.6684	0.6528	0.347	0.6714
NLLB-200	0.7861	0.7931	0.7912	0.221	0.8047
Google API	0.8902	0.9025	0.8907	0.1512	0.9386

4.3 BART-Based Automatic Labeling Analysis

Ground-truth sentiment labels were generated through a hybrid strategy combining expert human annotation (covering approximately 10% of the dataset) with BART-based zero-shot classification applied to the remaining samples. This design balances annotation reliability against the prohibitive cost of fully manual labeling at scale (~13,000 instances). The five-class sentiment scheme adopted Very Negative, Negative, Neutral, Positive, and Very Positive captures finer opinion gradations than conventional three-class

approaches, directly enabling the ordinal rating mapping central to inconsistency detection.

4.4 Training Performance Analysis

The training history of RoBERTa-large shows steady improvement over five epochs. Training loss decreased from 0.6704 in the first epoch to 0.0829 in the fifth, while accuracy increased from 0.8583 to 0.8901 and F1 rose from 0.8720 to 0.8948. Validation loss increased during training, which suggests a gap between training and validation behavior, but the final MAE and QWK values remained strong enough to show that the model

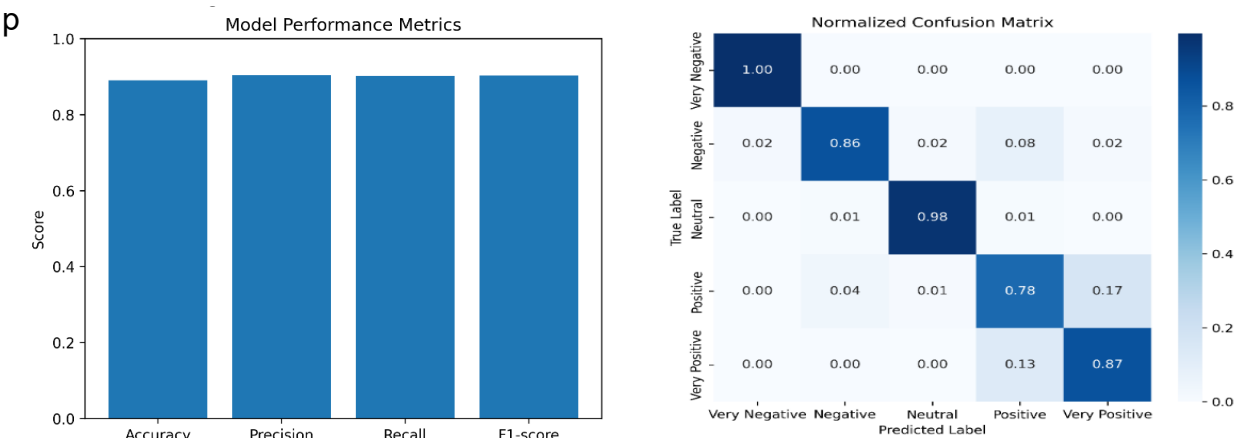


Fig. 5 Sentiment classification accuracy and confusion matrix

In fig. 5 shown the confusion matrix implies that the model is able to perform well in all five sentiment classes, with most predictions identify on the diagonal, that indicate the accurate classifications. The majority of the classification errors are made between neighbouring sentiment levels, in comparison with opposite sentiments (e.g., Positive vs. Very Positive, Negative vs. Positive). This indicates that the model can well capture the sentiment polarity but sometimes fails to differentiate the sentiment intensities. Overall, the confusion matrix shows that the classification performance is good for most of the sentiment classes and the classification is ordinal consistent between sentiment classes.

4.5. Comparative Analysis of Models

This large-scale comparison validates the benefits of transformer models over traditional baselines. Table 6. Shows that Logistic Regression was slightly better, with accuracy 0.8123 and QWK 0.8602, than SVM 0.8039 accuracy and 0.8483 QWK among the conventional methods. Transformer models achieved better results in the primary metrics. BART-base achieved 0.8375 accuracy, DeBERTa-v3-base reached 0.8774 accuracy with 0.9337 QWK, and RoBERTa-large produced the best overall result with 0.8902 accuracy, 0.90247 macro-F1, 0.893 weighted-F1, 0.1512 MAE, and 0.9386 QWK.

Table 6. Performance comparison of traditional and transformer-based models on the five-class sentiment classification task.

Paradigm	Models	Accuracy	Macro_f1	Weighted-F1	MAE	QWK
Traditional NLP	SVM	0.8039	0.8030	0.8043	0.3131	0.8483
	Logistic Regression	0.8123	0.8123	0.8138	0.2939	0.8602
Transformer NLP	BART-Base	0.8375	0.8350	0.8460	0.2540	0.8970
	DeBERTa-V3-Base	0.8774	0.8911	0.8530	0.1674	0.9337
	RoBERTa-Base	0.8697	0.8848	0.8900	0.1766	0.9282
	RoBERTa-Large	0.8902	0.9025	0.8930	0.1512	0.9386

The results are in line with the overall goal of the study, which is not only to classify sentiment but also produce ratings that are sensitive to the ordinal nature of customer feedback.

4.6 Sentiment-Driven Rating Analysis

Table 7 shows a sample of the review-level sentiment predictions made by the fine-tuned RoBERTa-Large model and the confidence scores for those predictions. The model was able to make an unambiguous discrimination between the sentiment categories with high confidence level, with the vast majority of samples categorized with a confidence greater than 0.999.

Table 7. Sample sentiment predictions scores generated by the RoBERTa-Large model.

Customer Review	True Sentiment	Predicted Sentiment
Absolutely love this dress! fits true to size and makes anyone look fabulous	Very Positive	Very Positive
The product does not match the description at all.	Very Negative	Very Negative
Would not recommend this Smartphones. The GalaxyWave S5 has poor portability and the display quality is frustrating.	Negative	Negative
The NoiseCancel Headphones serves its purpose. The build quality is okay and the battery life works fine most of the time.	Positive	Positive

The quality of the bag is good but the words and deeds did not match. They gave a different color.	Positive	Negative
Everything matched the description.	Neutral	Neutral
Fitting that the ultimate survivor of an epidemic is the one who will save her, the person who	Positive	Positive
Too big for cat food cans. Not worth it to send them back.	Negative	Negative

The single misclassification case displayed is a case that falls under a semantically complex category, where a positive review begins with a positive product attribute statement and concludes with a negative statement about product fulfillment (wrong color delivered). These reviews with mixed sentiments present a fundamental difficulty for sentiment classification using a single label and are a known problem with sentence-level sentiment models. Nevertheless, the model was able to effectively capture the dominant negative signal from the unmet expectation, which is arguably more useful for platform moderation.

4.7 Actual vs Sentiment Drive Product Rating Comparison

The differences between these ratings are presented in Table 8, which shows a strong positive skew in rating inconsistencies. This scatter plot (Fig. 6) illustrates the correlation between the original rating and the predicted rating, with a general upward trend, but also a significant variance in ratings around the identity line for a number of products with high ratings (4.5-5.0) indicating that a high rating on the platform does not necessarily mean a high rating from the reviews. The proposed framework achieved an MAE of 0.1512, showing that the sentiment-driven predicted ratings deviated only slightly from the reference rating values on average.

Table 8 Rating Differences Between Platform and Predicted Ratings

SKU	Product	Reviews	Platform Rating	Predicted Avg. Rating	Difference
2591265798	Sofa Set	21	5.0	3.0	+2.0 ▲
1544440888	Gown 1piece	22	5.0	3.5	+1.5 ▲
1253865069	Travel Bag	30	5.0	3.6	+1.4 ▲
2576500000	Abayas & Long Dresses	28	4.7	3.4	+1.3 ▲
2130255817	Bed Sheets	32	4.6	3.3	+1.3 ▲
2569548008	Non Stick Cookware Set	28	5.0	4.1	+0.9 ▲
2047544033	Women's Perfume	32	4.6	3.7	+0.9 ▲
1102200000	Women's Ear Ring	34	4.9	3.9	+1.0 ▲
2427500000	Arts & Crafts for Kids	22	3.8	4.1	-0.3 ▼
1818340187	Almirah	23	3.5	3.7	-0.2 ▼
2539700000	Baby Girls Shoes	25	4.2	4.2	0.0 ✓

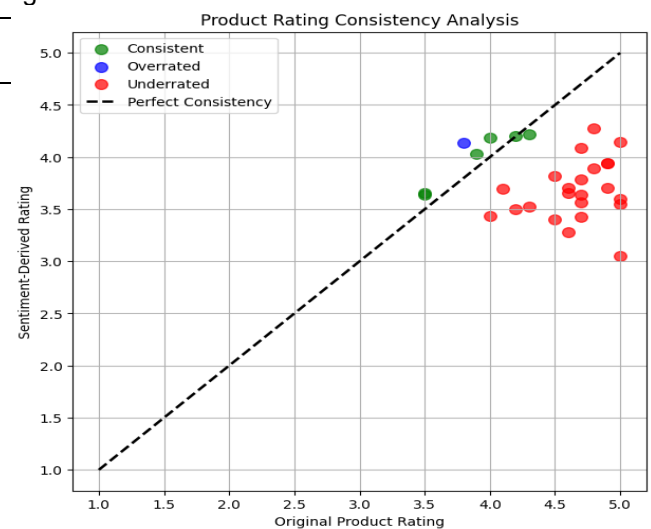


Fig. 6 Scatter Plot of Product Rating Consistency Analysis

4.8 Product-Level Rating Inconsistency

This is because the addition of the inconsistency detection is the main novelty of this work, and this discrepancy between manual and average rating based on sentiment analysis of customer reviews is the core subject of the work. Following Algorithm 1, products with $|D_p| \leq 0.3$ were treated as Consistent, products with $D_p > 0.3$ as Overrated, and products with $D_p < -0.3$ as Underrated. Table 6 presents the results of this comparison for 19 product SKUs from a variety of product categories which provide empirical evidence of systematic rating inflation in the evaluated e-commerce ecosystem.

As shown in Table 8, positive rating bias is present throughout the product catalog. The platform gave a sentiment-based predicted rating of 3.0, 3.5 and 3.6 for the Three Star rated Sofa Set, Gown and Travel Bag, respectively, which were off by 2.0, 1.5 and 1.4 stars. The gaps indicate that published ratings significantly over-report customer sentiment, a phenomenon commonly found in emerging-market e-commerce systems that are known to have had cases of review manipulation, including positive ratings in exchange for money or free products, and ratings that were solicited by the seller.

On the other hand, a small number of products (Arts & Crafts for Kids, Almirah) were rated lower on the platform than in the sentiment ratings, which may mean the platform ratings were not accurate enough, and the sentiment rating could be based on a greater volume of customer reviews. The difference in ratings for one product (Baby Girls Shoes, SKU 2539700000) was 0.0, indicating perfect consistency between displayed and sentiment inferred ratings for products that actually had a sentiment expressed in the reviews.

4.9 Error and Misclassification Analysis

Pattern I - Adjacent-Class Confusion: In table 9 show that the majority of errors were between the classes that are adjacent to one another such as between the Positive and Very Positive classes. The misclassifications are ordinal errors and fall within the range of the low MAE (0.1512), which means prediction errors tend to be within a single sentiment level. Hence, they have a restricted influence on the rating generation in the light of sentiment.

Table 9. Representative Misclassified Samples and Error Patterns

Serial No	Text	True_Sentiment	Predicted_Sentiment
3353	TV is good, delivery is super fast 24 hours wi...	Negative	Positive
10585	Having received my Fire TV today, I've been tr...	Very Positive	Positive

10520	Had older model, that you could text to speech...	Positive	Negative
2802	I am 32 years old and have been buying a lot o...	Negative	Positive
12948	These are the perfect shoe for a chubby footed...	Very Positive	Positive
3026	Super quality display and sound just awesome. ...	Positive	Very Positive
48	The laptop was good, there were no problems, i...	Very Positive	Positive
13700	I got the same product. I used the perfume yes...	Very Positive	Positive
11145	As with other recent Fire covers by Amazon, th...	Positive	Very Positive
2444	Great TV!!! Good service, Netflix, Youtube eve...	Positive	Very Positive

Pattern II – Mixed-Sentiment Reviews: Some of the errors observed in the mixed-sentiment reviews . These reviews send mixed signals to the public, resulting in some polarization changes. This restriction is typical of single-label sentiment classification and might be extended in future research with the aspect-level sentiment analysis approach. The error analysis reinforces the model's ability to make sense of misclassifications in terms of the meaning and that they have a limited, mostly harmless impact on the downstream task of rating prediction, thereby ensuring the operational reliability of the proposed framework.

5. Discussion

The proposed framework has three key differentiations from existing studies. First, it is not just sentiment analysis but rather also rating consistency detection. Second, it proposes a 5 class sentiment scheme to offer finer sentiment granularity for the rating prediction. Third, the framework is tailored for multilingual ecommerce reviews, such as Bangla, Banglish and mixed language reviews. The experimental results show that transformer models are powerful enough to produce sentiment-based ratings and detect mismatch between the sentiment of the review content and the ratings on the platform. The proposed RoBERTA-Large system is found to be the best among all the compared models as it has been able to tackle the most challenging combination of factors, namely 5-class sentiment classification over four language types (Bangla, Banglish, English, and Mixed) on 13,620 reviews.

Table 10. Comparison of the Proposed System with Related Work

Model	Language	Total Review s	Senti ment Class	Accurac y	Reference
Base LLM -Flan t5	Bangla,Banglish, English,Mixed	3208	3-class	89%	[13]
Fine Tuned Llama- 3.1 (8B)	Bengali and English	78,130	2class	95.5 %	[14]
Logistic Regression	English	2,05052	3-class	86%	[16]
Fine-Tuned LLAMA	Bangla	1,000	4-class	89.76%,	[28]
RoBERTa	English	22,641	3-class	72.47%	[29]
			5-class	87.69%	
Transformer- ensemble	Bangla	11,807	2-class	95.96%	[43]
LSTM	Bangla	1400	2-class	91.07%	[44]
RoBERTa-Large	Bangla, Banglish, English, Mixed	13,620	5-class	89.02	[Proposed System]

However, the accuracy figures 5. of the existing models are obtained on much more limited setups, including binary classification, single language datasets, or huge datasets for training. In Table 10 few prior models combine 5-class granularity with four language varieties at this scale. The resulting accuracy of 89.02% under these constraints suggests that the proposed system provides a practical and context-aware solution for sentiment analysis in the context of Bangladeshi multilingual texts or Banglish and Mixed-language content which is prevalent in the real world but not that much covered in the previous research.

6. Conclusion

This study accomplishes its primary objective by systematically investigating and resolving the inconsistencies between sentiment driven natural language reviews and numerical star ratings in e-commerce platforms. we discovered a significant discrepancy between the product ratings and the customer comment regarding any items. To build a bridge between product rating and review by A comprehensive dual experiment methodology was proposed to achieve this. First, a translation pipeline consisting of Google Translate API, NLLB-200, and M2M100 was implemented to process multilingual, Banglish, and code-mixed (Bangla-English) reviews. The translated data were subsequently used to build a customized dataset, with annotations generated using both human experts and pre-trained language models. Second, these refined inputs were used to train, test, and evaluate traditional NLP models (logistic regression and SVM) and advanced transformer NLP models (BART, DeBERTa, and RoBERTa). Find out different types of metrics, including accuracy, macro-F1, mean absolute error (MAE), and

quadratic-weighted kappa (QWK), showing the transformer NLP outperformed baseline models.

The novel contribution of this work is the development of a localized, sentiment-aware framework for processing complex, code-mixed reviews in e-commerce, which is evaluated using transformer models and compared to a task poorly handled by baseline NLP models. Which helps to show inconsistencies accurately. One major limitation of the proposed system is that the product rating is automatically generated from customer reviews but when a product has very few reviews or no reviews at all, the predicted rating may become biased and may incorrectly represent the actual quality of the product, either positively or negatively. In further study, work will extend the dataset to more regional languages and explore efficient LLMs fine-tune model to compare more result.

Declarations

1. Author Contributions

Pranta Kumar Sarkar: Conceptualization, Investigation, Data curation, Data collection, Methodology, Software, Validation, Formal analysis, Experiments, Results generation, Writing—original draft, and Visualization. **Kazi Aklima:** Writing—original draft (Abstract, Introduction, and Conclusion), Writing—review and editing. **Shahela Pervin:** Visualization, Figure preparation, Literature review, Methodology, and Writing—review and editing. **Mosqura Hoque:** Data collection, Data curation, Dataset description, and Writing—review and editing. All authors have read and agreed to the published version of the manuscript.

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3. Data Availability

Data Availability: The dataset used in this study is available from the corresponding author upon reasonable request.

4. Competing Interests

Competing Interests: The authors declare that they have no competing interests.

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